

(11.2) Najděte reálnou matici B takovou, aby platily rovnosti $ABA = A$, $BAB = B$ a zároveň AB i BA byly symetrické matice.

Najdeme inverzní matici k A

$$\left(\begin{array}{cccc|ccc} 1 & 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -2 & 1 & 0 & 1 & 0 \\ 1 & -2 & 1 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$= \left(\begin{array}{cccc|ccc} 1 & 1 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & -3 & 3 & -1 & 1 & 0 \\ 0 & -3 & 0 & 3 & -1 & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{cccc|ccc} 1 & 1 & 1 & -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -3 & 1/3 & 0 & -1/3 \\ 0 & 0 & 1 & -1 & 1/3 & -1/3 & 0 \end{array} \right)$$

$$\left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 2 & 1/3 & 1/3 & 1/3 \\ 0 & 1 & 0 & -3 & 1/3 & 0 & -1/3 \\ 0 & 0 & 1 & -1 & 1/3 & -1/3 & 0 \\ & & & & 0 & 0 & 0 \end{array} \right)$$

$$\left(\begin{array}{cccc|ccc} 1 & 1 & 1 & -2 & 1/3 & 1/3 & 1/3 \\ 1 & 1 & -2 & 1 & 1/3 & 0 & -1/3 \\ 1 & -2 & 1 & 1 & 1/3 & -1/3 & 0 \\ & & & & 0 & 0 & 0 \end{array} \right) = \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 1 & 0 & 1/3 & 0 & -1/3 \\ 0 & 0 & 1 & 1/3 & -1/3 & 0 \end{array} \right)$$

$$\left(\begin{array}{ccc|ccc} 1/3 & 1/3 & 1/3 & 1 & 1 & -2 \\ 1/3 & 0 & -1/3 & 1 & 1 & -2 \\ 1/3 & -1/3 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{array} \right) = \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & -2 \\ 0 & 1 & 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right)$$

$$B = \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 0 & -1/3 \\ 1/3 & -1/3 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$